EXPLORING CUSTOMER BEHAVIOR THROUGH DATABASE ANALYSIS

SQL / MS-SQL Server Project

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Database: OnlineFoodDelivery

Technology: SQL / MS-SQL Server

1. Project Overview

- This project explores customer behavior for an online food ordering platform using SQL.
- The analysis focuses on sales revenue, product performance, customer spending, visit frequency and popular products.
- Gold membership transactions are also analyzed where the required membership data is available.
- The objective is to convert transactional data into structured business insights.

2. Problem Statement & Objectives

- Problem: Understand customer visits, spending, favorite menu items and loyalty-program behavior.
- Analyze sales revenue generated by products.
- Identify top products based on revenue.
- Analyze customer spending and visit frequency.
- Identify first orders and most purchased products.
- Compare sales revenue across years.
- Generate actionable business insights.

3. Database & Tables

OnlineFoodDelivery database

Sales	Sales transactions	userid, created_date, product_id
Product	Product details	product_id, product_name, price
User_name	Customer names	userid, Names
Users	Customer signup data	userid, signup_date
Goldusers_Signup	Gold membership data	user_id, gold_signup_date

4. Methodology

- Created the OnlineFoodDelivery database in MS-SQL Server.
- Created the required relational tables and inserted the records available in the project material.
- Used JOIN operations to combine related customer, product and transaction data.
- Used GROUP BY, COUNT and SUM for summary analysis.
- Used CTEs and Window Functions for advanced analysis.
- Used YEAR, MIN and DATEDIFF for date-based analysis.
- Used TOP, RANK, ROW_NUMBER and LAG where required.

5. SQL Analysis – Q1 to Q5

- Q1. Total sales revenue per product.
- Q2. Top 3 products by sales revenue.
- Q3. Number of users who signed up and took Gold membership.
- Q4. Revenue generated from Gold users.
- Q5. Total revenue from Gold users.

6. SQL Analysis – Q6 to Q10

- Q6. How long each user has been a Gold user.
- Q7. Most popular product among Gold users.
- Q8. Total sales revenue each year.
- Q9. Sales revenue change over years.
- Q10. Average Gold signup compared to just signup.

7. SQL Analysis – Q11 to Q15

- Q11. Number of orders by Gold member users.
- Q12. Total amount spent by each customer.
- Q13. Customer visit frequency.
- Q14. First order by each customer.
- Q15. Most purchased item and count.

8. SQL Analysis – Q16 to Q20

- Q16. Most popular item for each customer.
- Q17. First item purchased after becoming Gold.
- Q18. Item purchased before becoming Gold.
- Q19. Total orders and amount spent before becoming Gold.
- Q20. Rank all transactions for each member while Gold.

9. SQL Concepts Used

- SELECT, JOIN and GROUP BY
- Subqueries and Common Table Expressions (CTEs)
- LAG() for year-over-year comparisons
- Relational database concepts and date-based analysis

- COUNT(), SUM(), MIN(), YEAR() and DATEDIFF()
- RANK() and ROW_NUMBER()
- TOP for top-performing products and results

10. Key Insights & Recommendations

- Compare product revenue to identify stronger-performing products.
- Use customer spending to identify and retain higher-value customers.
- Use visit frequency to understand customer engagement.
- Use popular-product analysis for menu and promotion planning.
- Use year-wise sales trends for business planning.
- Evaluate Gold membership using complete Gold-user transaction data.
- Promote strong products and use customer behavior for targeted engagement.

11. Conclusion & Project Limitation

- SQL and MS-SQL Server provide a structured way to analyze customer behavior from transaction, product and customer data.
- The project demonstrates analysis of spending, visits, products, sales trends and loyalty-program behavior.
- The project material states 62 Sales records and 5 Gold members.
- Complete readable Goldusers_Signup records were not available in the supplied material used for this implementation.
- Therefore, Gold-membership results requiring missing rows should not be presented as complete numerical findings.

THANK YOU

Exploring Customer Behavior Through Database Analysis

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